10. The female sex hormone is:

(a) Estrogen (b) Androgen
(c) Insulin (d) Oxytocin
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

Among the given options, estrogen (oestrogen) is the female sex hormone. It is produced primarily by the ovaries and responsible for the development and function of the female reproductive organs and sexual characteristics.

11. Which gland provides both endocrine and exocrine function?

(a) Thyroid (b) Adrenal
(c) Pancreas (d) More than one of the above
(e) None of the above

B.P.S.C. Headmaster 07-12-2023

Ans. (c)

Pancreas is mixed gland which provide both endocrine and exocrine function. In exocrine function pancreas secretes enzymes to break down the proteins, lipids and carbohydrates in food. Functioning as an endocrine gland, the pancreas secretes the hormones insulin and glucagon to control blood sugar levels throughout the day.

VIII. Reproduction and Embryo Development

1. The entry of pollen tube into the ovule through micropyle is called

(a) chalazogamy (b) mesogamy
(c) anisogamy (d) porogamy

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (d)

The entry of Pollen tube into the Ovule through micropyle is called Porogamy. Mesogamy refers to a condition when Pollen Tubes enters via the integuments. Anisogamy is the form of sexual reproduction that involves the union or fusion of two gametes which differ in size and/or form.

2. Where does fertilization occur?

(a) In uterus (b) In oviduct
(c) In ovary (d) In vagina

45th B.P.S.C. (Pre) 2001

Ans. (b)

Fertilization occurs at the end of the fallopian tube or oviduct away from the uterus (close to the ovary). The oviduct is the tube that links ovary to the uterus and ovulated oocyte travels down to become fertilized by sperm present in the female tract.

3. Which of the functions is performed by the ovaries?

(a) Formation of ovum (b) Secretion of progesterone
(c) Secretion of estrogen (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (d)

Ovaries play a critical role in both menstruation and conception. They produce eggs for fertilization and they make the hormones estrogen and progesterone.

4. Which of the following is not a part of the female reproductive system in human beings?

(a) Ovary (b) Uterus
(c) Vas deferens (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (c)

The vas deferens are not part of the female reproductive system in humans. The vas deferens is a duct that carries sperm and is part of the male reproductive system. The female reproductive system includes Ovaries, Oviducts, Uterus, Cervix Vagina etc.

5. Asexual reproduction takes place through budding in ______

(a) Amoeba (b) Yeast
(c) Plasmodium (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (b)

Asexual reproduction takes place through budding in yeast. In budding, a small knob or bud forms on the parent cell grows, and then separates to become a new yeast cell. The parent cell develops tiny protrusions before developing into a bud.

PLANT PHYSIOLOGY

I. Photosynthesis

1. The stomata open or close due to change in the

(a) position of nucleus in cells
(b) protein composition of cells
(c) amount of water in cells
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)